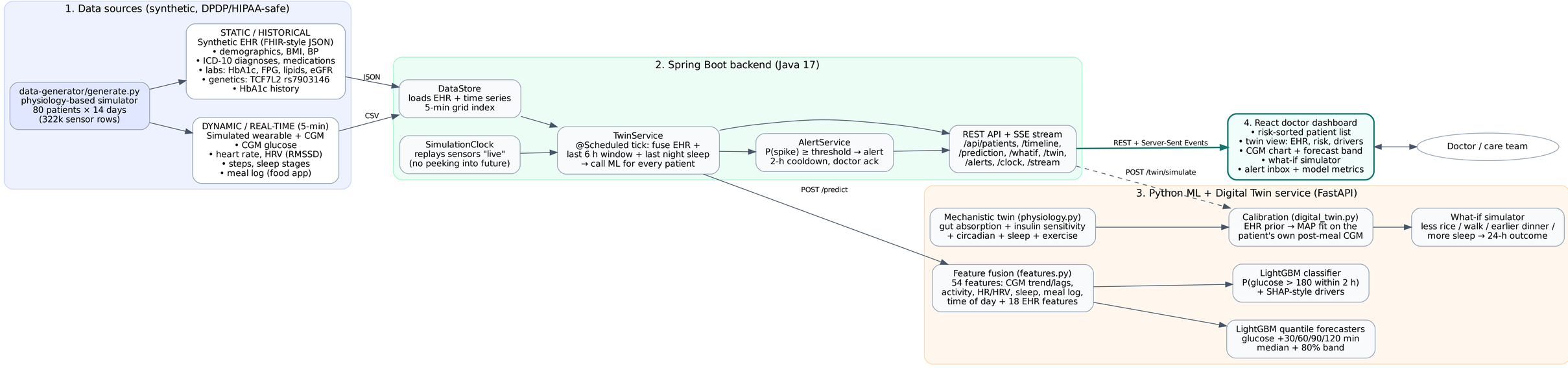
GlucoTwin — Digital Twin Architecture (Type 2 Diabetes, 2-hour glucose-spike prediction)
1. Data sources (synthetic, DPDP/HIPAA-safe)
data-generator/generate.py
physiology-based simulator
80 patients × 14 days
(322k sensor rows)
STATIC / HISTORICAL
Synthetic EHR (FHIR-style JSON)
• demographics, BMI, BP
• ICD-10 diagnoses, medications
• labs: HbA1c, FPG, lipids, eGFR
• genetics: TCF7L2 rs7903146
• HbA1c history
DYNAMIC / REAL-TIME (5-min)
Simulated wearable + CGM
• CGM glucose
• heart rate, HRV (RMSSD)
• steps, sleep stages
• meal log (food app)
JSON
CSV
2. Spring Boot backend (Java 17)
DataStore
loads EHR + time series
5-min grid index
SimulationClock
replays sensors "live"
(no peeking into future)
TwinService
@Scheduled tick: fuse EHR +
last 6 h window + last night sleep
→ call ML for every patient
AlertService
P(spike) ≥ threshold → alert
2-h cooldown, doctor ack
REST API + SSE stream
/api/patients, /timeline,
/prediction, /whatif, /twin,
/alerts, /clock, /stream
POST /predict
REST + Server-Sent Events
POST /twin/simulate
4. React doctor dashboard
• risk-sorted patient list
• twin view: EHR, risk, drivers
• CGM chart + forecast band
• what-if simulator
• alert inbox + model metrics
Doctor / care team
3. Python ML + Digital Twin service (FastAPI)
Mechanistic twin (physiology.py)
gut absorption + insulin sensitivity
+ circadian + sleep + exercise
Calibration (digital_twin.py)
EHR prior → MAP fit on the
patient's own post-meal CGM
What-if simulator
less rice / walk / earlier dinner /
more sleep → 24-h outcome
Feature fusion (features.py)
54 features: CGM trend/lags,
activity, HR/HRV, sleep, meal log,
time of day + 18 EHR features
LightGBM classifier
P(glucose > 180 within 2 h)
+ SHAP-style drivers
LightGBM quantile forecasters
glucose +30/60/90/120 min
median + 80% band